

Deploying ActiveGate

Series: ONBRD | **Notebook:** 3 of 10 | **Created:** December 2025 | **Last Updated:** 01/28/2026

Your Network Gateway to Dynatrace

ActiveGate is a lightweight component that routes traffic between your infrastructure and Dynatrace. This notebook covers when you need ActiveGate, how many to deploy, where to place them, and installation steps - including comprehensive Kubernetes deployment options.

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Prerequisites

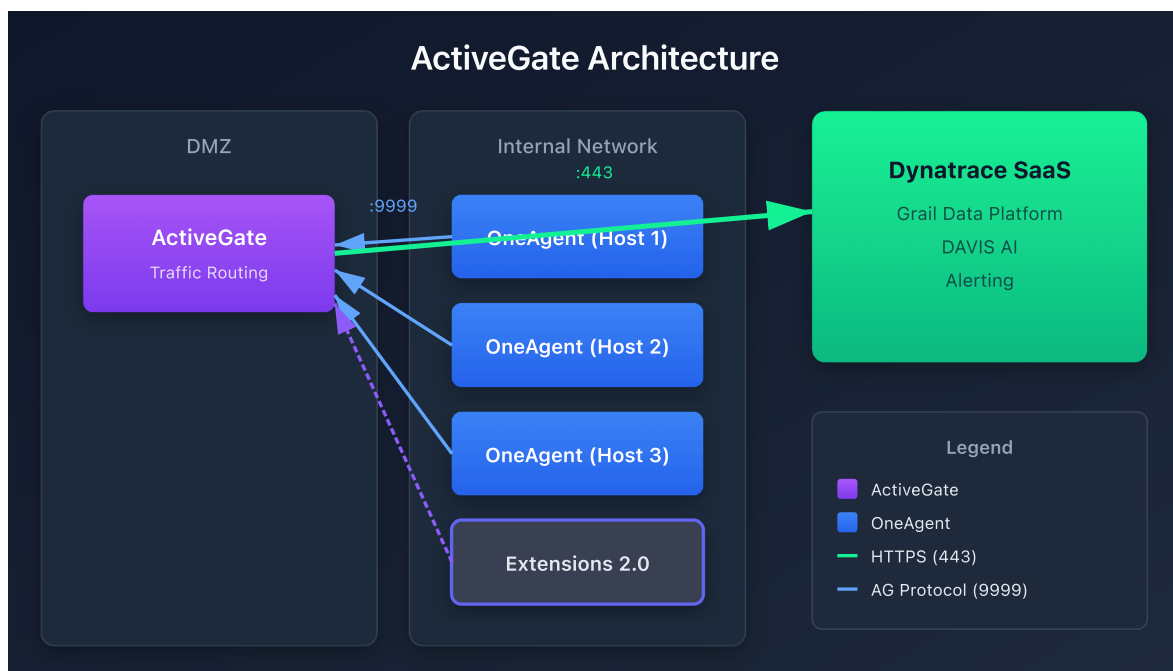
- Admin access to Dynatrace environment
- Network architecture diagram (know your zones)
- Server for ActiveGate (Linux, Windows, or Kubernetes cluster)
- Outbound HTTPS (443) to Dynatrace SaaS

1. What is ActiveGate?

ActiveGate is a proxy and routing component that connects your environment to Dynatrace.

Component	Purpose
Environment ActiveGate	Routes OneAgent traffic, runs extensions, executes synthetics
Cluster ActiveGate	Dynatrace Managed only - cluster communication

Note: This notebook covers Environment ActiveGate for Dynatrace SaaS.



ActiveGate Capabilities

Capability	Description
Traffic Routing	Proxies OneAgent data to Dynatrace
Extensions 2.0	Runs Python-based extensions locally
Synthetic Monitoring	Private synthetic locations
Cloud Integrations	AWS, Azure, GCP metric collection
Kubernetes API	Cluster monitoring via API
Log Ingest	Generic log ingest endpoint

2. When Do You Need ActiveGate?

Required Scenarios

Scenario	Why ActiveGate is Required
Network-restricted hosts	OneAgents can't reach internet directly
Private synthetic monitors	Test internal applications
Extensions 2.0	Custom data sources (SNMP, databases, etc.)
AWS/Azure/GCP monitoring	Pull cloud metrics via API
VMware monitoring	vCenter integration
Kubernetes full-stack	Cluster API access for events, metrics

Optional but Recommended

Scenario	Benefit
Large deployments (500+ hosts)	Reduces outbound connections
Multiple network zones	Centralized routing per zone
Bandwidth optimization	Compresses and batches data
Security compliance	Single egress point for audit

Decision Tree



3. How Many ActiveGates?

Sizing Guidelines

ActiveGate Size	OneAgents Supported	Memory	CPU
Small	Up to 500	2 GB	2 cores
Medium	500-1,500	4 GB	4 cores
Large	1,500-3,000	8 GB	8 cores
Extra Large	3,000-5,000	16 GB	16 cores

Detailed Hardware Requirements

Minimum System Requirements

Component	Minimum	Recommended
CPU	2 cores (x64)	4+ cores
RAM	2 GB	4+ GB
Disk Space	2 GB	10+ GB (for logs/cache)
Disk Type	HDD	SSD (for extensions)
Network	100 Mbps	1 Gbps

Operating System Support

OS	Supported Versions
RHEL/CentOS	7.x, 8.x, 9.x
Ubuntu	18.04, 20.04, 22.04, 24.04 LTS
Debian	10, 11, 12
Amazon Linux	2, 2023
SUSE	12 SP5, 15 SPx
Windows Server	2016, 2019, 2022

Memory Sizing by Workload

Workload Type	Base Memory	Additional per Feature
Routing Only	2 GB	-
+ Extensions 2.0	2 GB	+1-2 GB per active extension
+ Synthetic	2 GB	+1 GB per 10 monitors
+ Cloud Integration	2 GB	+512 MB per cloud account
+ Log Ingest	2 GB	+2 GB for high volume

Disk Space Sizing

Purpose	Space Required
Base Installation	1.5 GB
Log Files (default)	1-5 GB (rotated)
Extension Cache	500 MB - 2 GB
Synthetic Cache	500 MB - 1 GB
Temporary Files	1 GB
Recommended Total	10-20 GB

Best Practice: Mount `/opt/dynatrace` on a dedicated partition with at least 20 GB for production deployments.

High Availability

For production environments, deploy **at least 2 ActiveGates per network zone**:

Deployment	ActiveGates	Purpose
Minimum HA	2 per zone	Failover capability
Recommended	2-3 per zone	Load distribution + failover
Large scale	N+1 per zone	Capacity + redundancy

Key Point: OneAgents automatically load-balance across available ActiveGates and failover if one becomes unavailable.

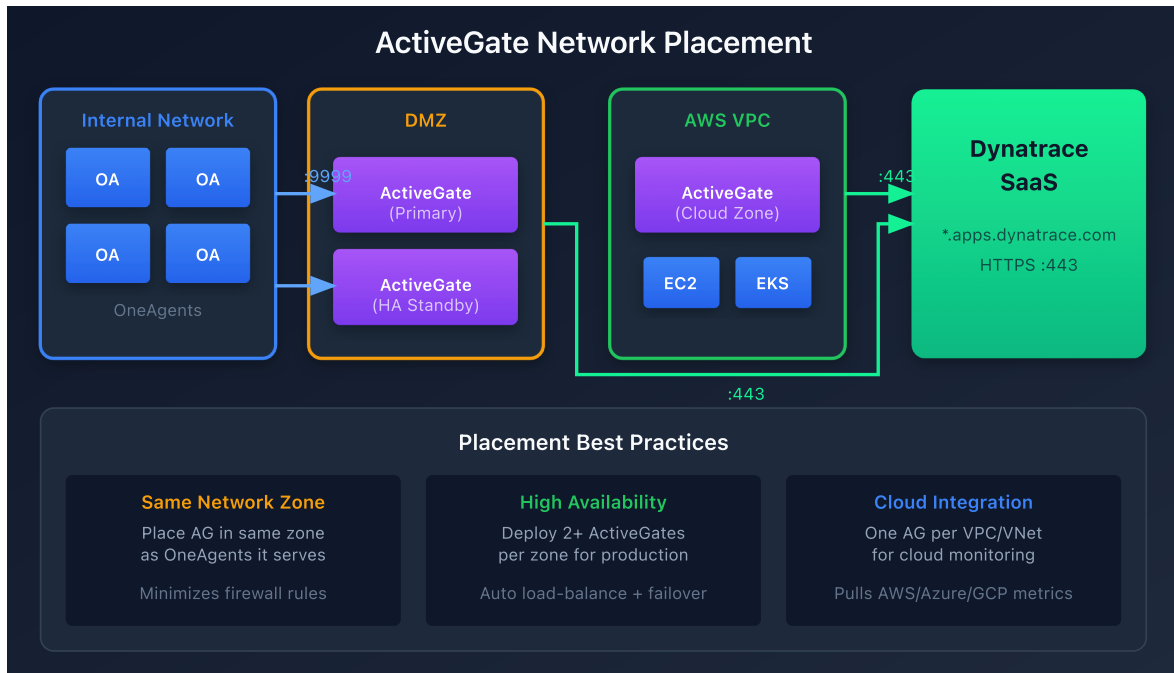
Example Deployment

Network Zone	Hosts	ActiveGates	Sizing	Memory	CPU
Production DMZ	200	2	Small	2 GB each	2 cores
Production Internal	800	2	Medium	4 GB each	4 cores
Development	150	1	Small	2 GB	2 cores
Total	1,150	5		14 GB	14 cores

4. Where to Deploy?

Network Zone Strategy

Deploy ActiveGates based on network segmentation:



Placement Rules

Rule	Description
Same network zone	AG should be in same zone as OneAgents it serves
Outbound access	AG needs HTTPS to *.dynatrace.com
Inbound from agents	OneAgents connect to AG on port 9999
Low latency	Place AG close to monitored workloads

Cloud-Specific Guidance

Cloud	Recommendation
AWS	Deploy in management VPC or shared services subnet
Azure	Hub VNet with peering to spoke VNets
GCP	Shared VPC host project
Kubernetes	Deploy as StatefulSet or standalone VM outside cluster

5. Generating Tokens

ActiveGate requires a PaaS token for installation.

Location: Settings → Integration → Platform as a Service

Or use the API token approach:

Required Token Scopes

Scope	API Name	Purpose
PaaS integration - Installer download	<code>InstallerDownload</code>	Download ActiveGate installer

Additional Scopes by Use Case

Use Case	Additional Scopes
AWS Monitoring	<code>aws.supportedServicesRead</code>
Azure Monitoring	<code>azure.supportedServicesRead</code>
Extensions 2.0	<code>extensions.read</code> , <code>extensions.write</code>
Synthetic Private Location	<code>syntheticLocations.write</code>

Token Naming Convention

Use descriptive names:

- `prod-activegate-dmz`
- `aws-activegate-useast1`
- `k8s-activegate-cluster1`

6. Installation Methods

Linux Installation

```
# Download the installer
wget -O Dynatrace-ActiveGate.sh \
  "https://{tenant-id}.live.dynatrace.com/api/v1/deployment/installer/gateway/unix/latest?Api-Token={paas-token}"

# Make executable and run
chmod +x Dynatrace-ActiveGate.sh
sudo ./Dynatrace-ActiveGate.sh
```

Windows Installation

```
# Download the installer
Invoke-WebRequest -Uri "https://{tenant-id}.live.dynatrace.com/api/v1/deployment/installer/gateway/windows/latest?Api-Token={paas-token}" -OutFile Dynatrace-ActiveGate.exe

# Run the installer
.\Dynatrace-ActiveGate.exe
```

Container Deployment (Docker/Podman)

```
docker run -d --name dynatrace-activegate \
  -e DT_TENANT="{tenant-id}" \
  -e DT_API_TOKEN="{paas-token}" \
  -p 9999:9999 \
  dynatrace/activegate:latest
```

Installation Parameters

Parameter	Purpose	Example
<code>--set-network-zone</code>	Assign to network zone	<code>--set-network-zone=dmz</code>
<code>--set-group</code>	Group for management	<code>--set-group=production</code>
<code>--enable-synthetic</code>	Enable synthetic capability	<code>--enable-synthetic</code>

6a. Kubernetes Deployment (Detailed)

Deploying ActiveGate in Kubernetes requires careful consideration of where, how, and when to use containerized ActiveGates vs. traditional VM deployments.

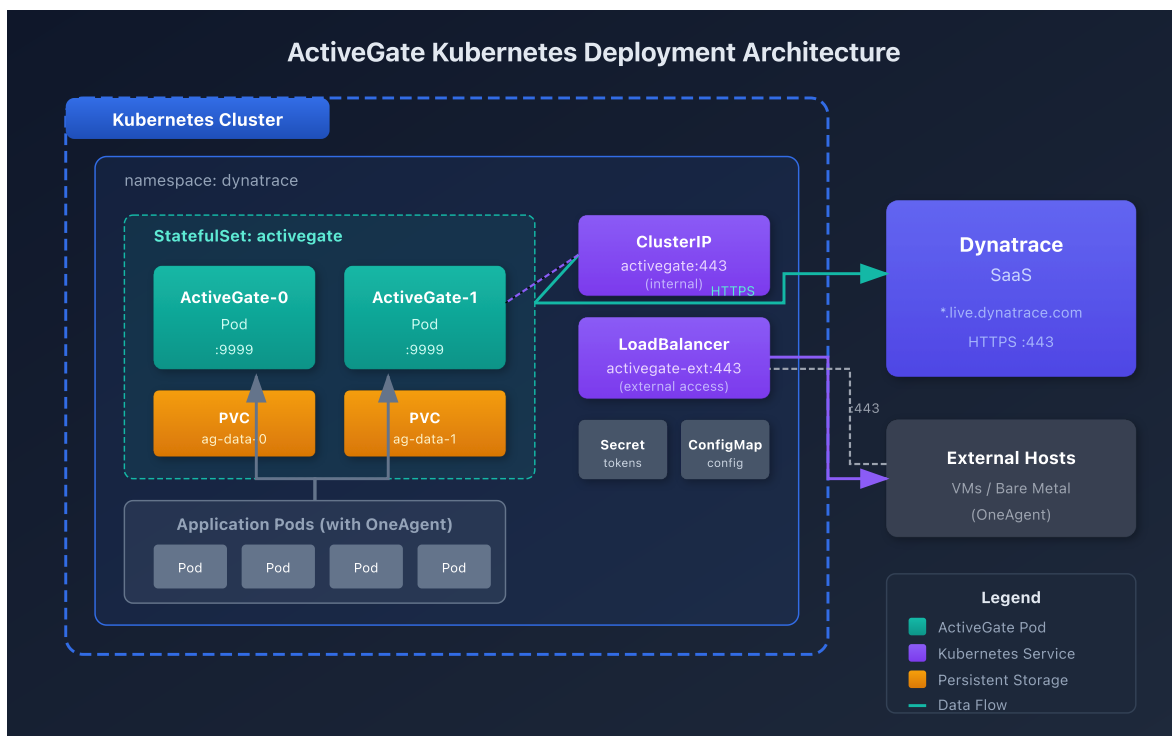
When to Deploy ActiveGate in Kubernetes

Scenario	Deploy in K8s?	Reasoning
Cluster-only monitoring	✅ Yes	Co-located with workloads, simplifies networking
Routing for in-cluster OneAgents	✅ Yes	Lower latency, no external hops
Extensions 2.0 for K8s resources	✅ Yes	Direct access to cluster APIs
Multi-cluster routing	⚠️ Consider	May need external AG for cross-cluster
Private synthetic monitoring	❌ Usually VM	Synthetic requires stable network, not pod restarts
Routing for external hosts	❌ VM preferred	External hosts need stable, routable IP
Air-gapped environments	❌ VM preferred	Often need dedicated proxy tier

When NOT to Deploy in Kubernetes

Scenario	Why VM is Better
Routing for VMs/bare metal	VMs need stable IPs, not pod IPs
Private synthetic locations	Synthetic tests sensitive to pod restarts
Extensions polling external systems	External systems may not allow K8s egress IPs
Compliance requiring dedicated hosts	Some compliance requires dedicated hardware

Deployment Architecture



Kubernetes Hardware Requirements

Deployment Size	CPU Request	CPU Limit	Memory Request	Memory Limit
Small (≤ 500 agents)	500m	2000m	1Gi	2Gi
Medium (500-1500)	1000m	4000m	2Gi	4Gi
Large (1500-3000)	2000m	8000m	4Gi	8Gi
Extra Large (3000+)	4000m	16000m	8Gi	16Gi

Method 1: Dynatrace Operator (Recommended)

The Dynatrace Operator manages ActiveGate lifecycle automatically.

Important: Use `apiVersion: dynatrace.com/v1beta5` for Dynatrace Operator 1.7.0+. Earlier versions (v1beta1, v1beta2) are deprecated and no longer supported.

```
# dynakube.yaml – ActiveGate via Operator
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # ActiveGate configuration
  activeGate:
    # Capabilities to enable
    capabilities:
      - routing          # Route OneAgent traffic
      - kubernetes-monitoring # K8s API monitoring
      - dynatrace-api     # Proxy Dynatrace API calls

  # Resource sizing
  resources:
    requests:
      cpu: "500m"
      memory: "1Gi"
    limits:
      cpu: "2000m"
      memory: "2Gi"

  # Number of replicas (HA)
  replicas: 2

  # Tolerations for dedicated nodes
  tolerations:
    - key: "dedicated"
      operator: "Equal"
      value: "dynatrace"
      effect: "NoSchedule"

  # Node selector (optional)
  nodeSelector:
    node-type: monitoring

  # Custom properties
  customProperties:
    value: |
      [connectivity]
      networkZone=kubernetes

  # Labels
```

```
labels:
  app.kubernetes.io/component: activegate
```

Install with Operator:

```
# Create namespace
kubectl create namespace dynatrace

# Create secret with tokens
kubectl -n dynatrace create secret generic dynakube \
  --from-literal=apiToken=<your-api-token> \
  --from-literal=paasToken=<your-paas-token>

# Install Dynatrace Operator via Helm
helm repo add dynatrace
https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable
helm repo update

helm install dynatrace-operator dynatrace/dynatrace-operator \
  --namespace dynatrace \
  --set installCRD=true

# Apply DynaKube configuration
kubectl apply -f dynakube.yaml
```

Method 2: Helm Chart (Standalone)

For more control, use the standalone ActiveGate Helm chart:

```
# Add Dynatrace Helm repo
helm repo add dynatrace
https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable
helm repo update

# Create values file
cat > activegate-values.yaml <<< 'EOF'
apiUrl: "https://{tenant-id}.live.dynatrace.com/api"

activeGate:
  replicas: 2

  capabilities:
    - routing
    - kubernetes-monitoring

  resources:
    requests:
      cpu: 500m
      memory: 1Gi
    limits:
      cpu: 2
      memory: 2Gi

# Persistent storage for logs
persistence:
  enabled: true
  size: 10Gi
  storageClassName: gp3

# Service configuration
service:
  type: ClusterIP # or LoadBalancer for external access

# Environment variables
env:
  - name: DT_NETWORK_ZONE
    value: "kubernetes"

# Token from existing secret
existingSecret: dynatrace-tokens
EOF

# Install
helm install activegate dynatrace/dynatrace-activegate \
```



```
--namespace dynatrace \  
--values activegate-values.yaml
```

Method 3: Raw Kubernetes Manifests

For complete control, use raw manifests:

```

# activegate-namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dynatrace
---
# activegate-secret.yaml
apiVersion: v1
kind: Secret
metadata:
  name: dynatrace-tokens
  namespace: dynatrace
type: Opaque
stringData:
  apiToken: "<your-api-token>"
  paasToken: "<your-paas-token>"
---
# activegate-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: activegate-config
  namespace: dynatrace
data:
  custom.properties: |
    [connectivity]
    networkZone=kubernetes

    [collector]
    MaxIncomingConnections=2000
---
# activegate-statefulset.yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: activegate
  namespace: dynatrace
  labels:
    app: activegate
spec:
  serviceName: activegate
  replicas: 2
  selector:
    matchLabels:
      app: activegate
  template:
    metadata:

```

```
labels:
  app: activegate
spec:
  serviceAccountName: dynatrace-activegate
  containers:
    - name: activegate
      image: dynatrace/activegate:latest
      imagePullPolicy: Always
      env:
        - name: DT_TENANT
          value: "{tenant-id}"
        - name: DT_API_TOKEN
          valueFrom:
            secretKeyRef:
              name: dynatrace-tokens
              key: paasToken
        - name: DT_CAPABILITIES
          value: "routing,kubernetes_monitoring"
      ports:
        - containerPort: 9999
          name: ag-https
      resources:
        requests:
          cpu: "500m"
          memory: "1Gi"
        limits:
          cpu: "2000m"
          memory: "2Gi"
      volumeMounts:
        - name: config
          mountPath: /var/lib/dynatrace/gateway/config/custom.properties
          subPath: custom.properties
        - name: ag-data
          mountPath: /var/lib/dynatrace/gateway
  livenessProbe:
    httpGet:
      path: /rest/health
      port: 9999
      scheme: HTTPS
    initialDelaySeconds: 30
    periodSeconds: 15
  readinessProbe:
    httpGet:
      path: /rest/health
      port: 9999
      scheme: HTTPS
    initialDelaySeconds: 30
    periodSeconds: 5
```

```

    volumes:
      - name: config
        configMap:
          name: activegate-config
    volumeClaimTemplates:
      - metadata:
          name: ag-data
        spec:
          accessModes: ["ReadWriteOnce"]
          storageClassName: gp3
          resources:
            requests:
              storage: 10Gi
---
# activegate-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
spec:
  type: ClusterIP # Change to LoadBalancer for external access
  selector:
    app: activegate
  ports:
    - port: 443
      targetPort: 9999
      name: https
---
# Headless service for StatefulSet DNS
apiVersion: v1
kind: Service
metadata:
  name: activegate-headless
  namespace: dynatrace
spec:
  clusterIP: None
  selector:
    app: activegate
  ports:
    - port: 9999
      name: ag-https

```

Exposing ActiveGate for External Access

If external hosts need to route through the K8s ActiveGate:

```
# Option 1: LoadBalancer Service
apiVersion: v1
kind: Service
metadata:
  name: activegate-external
  namespace: dynatrace
  annotations:
    # AWS NLB
    service.beta.kubernetes.io/aws-load-balancer-type: "nlb"
    service.beta.kubernetes.io/aws-load-balancer-scheme: "internal"
spec:
  type: LoadBalancer
  selector:
    app: activegate
  ports:
    - port: 443
      targetPort: 9999
---
# Option 2: Ingress (with TLS passthrough)
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: activegate-ingress
  namespace: dynatrace
  annotations:
    nginx.ingress.kubernetes.io/ssl-passthrough: "true"
    nginx.ingress.kubernetes.io/backend-protocol: "HTTPS"
spec:
  ingressClassName: nginx
  rules:
    - host: activegate.internal.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: activegate
                port:
                  number: 443
```

Platform-Specific Configurations

Amazon EKS

```
# EKS-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    service.beta.kubernetes.io/aws-load-balancer-type: "nlb"
    service.beta.kubernetes.io/aws-load-balancer-internal: "true"
    service.beta.kubernetes.io/aws-load-balancer-cross-zone-load-balancing-
enabled: "true"
spec:
  type: LoadBalancer
  # ...
```

Azure AKS

```
# AKS-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    service.beta.kubernetes.io/azure-load-balancer-internal: "true"
spec:
  type: LoadBalancer
  # ...
```

Google GKE

```
# GKE-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    cloud.google.com/load-balancer-type: "Internal"
spec:
  type: LoadBalancer
  # ...
```

High Availability in Kubernetes

Configuration	Setting
Replicas	2-3 for production
Pod Anti-Affinity	Spread across nodes
PodDisruptionBudget	minAvailable: 1
Resource Requests	Guarantee scheduling

```
# Pod Anti-Affinity for HA
spec:
  affinity:
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        - labelSelector:
            matchLabels:
              app: activegate
          topologyKey: kubernetes.io/hostname
---
# PodDisruptionBudget
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: activegate-pdb
  namespace: dynatrace
spec:
  minAvailable: 1
  selector:
    matchLabels:
      app: activegate
```

7. Verifying Deployment

After installation, verify ActiveGate is connected and healthy.

```
// ActiveGate verification – use the Entities API v2 or Settings API
// Note: ActiveGates are not directly queryable via DQL fetch
// Use: GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")
// Or check the Deployment Status → ActiveGates page in the Dynatrace UI
```

```
// ActiveGate version check – use the Entities API v2
// GET /api/v2/entities?
entitySelector=type("ENVIRONMENT_ACTIVE_GATE")&fields=properties
// The response includes version information in the properties
// Or check Deployment Status → ActiveGates in the UI for version details
```

```
// ActiveGate count – use the Entities API v2
// GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")
// Count the entities in the response, or check the ActiveGates page in UI
```


Verification via UI

Location: Deployment status → ActiveGates

Check for:

-  Status: Connected
-  Version: Latest or recent
-  Modules: Expected capabilities enabled

Verification Commands

Linux:

```
# Check service status
sudo systemctl status dynatracegateway

# Check connectivity
curl -k https://localhost:9999/communication/health

# View logs
sudo tail -100 /var/log/dynatrace/gateway/gateway.log
```

Windows:

```
# Check service status
Get-Service -Name "Dynatrace ActiveGate"

# Check connectivity
Invoke-WebRequest -Uri "https://localhost:9999/communication/health" -
SkipCertificateCheck
```

8. Troubleshooting

Common Issues

Issue	Cause	Solution
Not appearing in UI	Network blocked	Check firewall for 443 outbound
Shows disconnected	Service stopped	Restart dynatracegateway service
OneAgents not routing	Wrong network zone	Verify zone configuration
High memory usage	Too many agents	Add another AG or increase RAM
Certificate errors	Self-signed cert issue	Trust Dynatrace CA or use custom cert

Network Verification

```
# Test outbound to Dynatrace
curl -v https://{tenant-id}.live.dynatrace.com/communication/health

# Test ActiveGate is listening
netstat -tlnp | grep 9999

# Test from OneAgent host
curl -k https://{activegate-ip}:9999/communication/health
```

Log Locations

OS	Log Path
Linux	/var/log/dynatrace/gateway/
Windows	C:\ProgramData\dynatrace\gateway\log\

9. Next Steps

With ActiveGate deployed:

1. **ONBRD-04: Deploying OneAgent** - Now deploy agents that route through ActiveGate
2. Configure network zones if using multiple ActiveGates
3. Enable additional modules (synthetic, extensions) as needed
4. Set up monitoring for ActiveGate itself

Deployment Checklist

Traditional (VM/Bare Metal)

- ☐ ActiveGate deployed in each required network zone
- ☐ At least 2 per zone for HA (production)
- ☐ Status shows "Connected" in Dynatrace
- ☐ Firewall rules configured (443 out, 9999 in)
- ☐ Network zones configured (if applicable)
- ☐ Sizing appropriate for expected OneAgent count

Kubernetes

- ☐ Dynatrace Operator installed (recommended)
- ☐ DynaKube CR applied with activeGate configuration
- ☐ Replicas set to 2+ for production HA
- ☐ Resource requests/limits configured
- ☐ PodDisruptionBudget created
- ☐ Pod anti-affinity configured for spread across nodes
- ☐ Service type appropriate (ClusterIP vs LoadBalancer)

Summary

In this notebook, you learned:

- What ActiveGate does and its capabilities
- When ActiveGate is required vs. optional
- Hardware requirements and sizing guidelines

- How to size and plan ActiveGate deployment
 - Where to place ActiveGates in your network
 - Installation methods for Linux, Windows, and containers
 - **Kubernetes deployment** using Operator, Helm, or raw manifests
 - Platform-specific configurations for EKS, AKS, and GKE
 - How to verify successful deployment
-

References

General

- [ActiveGate Overview](#)
- [ActiveGate Installation](#)
- [Network Zones](#)
- [ActiveGate Sizing](#)
- [Private Synthetic Locations](#)

Kubernetes

- [Dynatrace Operator](#)
 - [Dynatrace Operator GitHub](#)
 - [ActiveGate on Kubernetes](#)
 - [DynaKube Custom Resource](#)
 - [Helm Chart Repository](#)
-

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